

**Challenge & Solution:** Campus recruitment portals evaluate candidate batches against single job postings. Standard keyword matching fails on terminology synonyms (e.g. *Express* vs *Node.js*), while generative cloud LLMs produce hallucinations, privacy issues, and arbitrary scores. **Brocode's Smart Shortlisting Engine** is a **100% offline, privacy-first recruitment platform** pairing deterministic lexical matching with offline dense vector search (all-MiniLM-L6-v2) and an **Evidence-First Explainability Engine** that provides auditable quotes for recruiter verification.

**Hard Competition Constraints Satisfied:**

- **Zero Cloud APIs:** No OpenAI, Gemini, Claude, or internet calls (HF\_HUB\_OFFLINE=1).
- **Non-Arbitrary Scoring:** Fully mathematical formulation combining lexical (\$L\$) and semantic (\$S\$) signals.
- **Format Resilient:** Robustly parses messy multi-column PDFs and isolates corrupt files without crashing.

1. End-to-End System Architecture (7 Pipeline Stages)

| Stage             | Component & Technology        | Core Functionality & Guarantee                                                                             |
|-------------------|-------------------------------|------------------------------------------------------------------------------------------------------------|
| 1. Ingestion      | PyMuPDF (fitz) + Normalizer   | Extracts text streams, detects multi-column layouts, normalizes whitespace & unicode.                      |
| 2. Intelligence   | JD & Resume Profilers         | Classifies mandatory (REQUIRED) vs bonus (PREFERRED) requirements; structures candidate profile.           |
| 3. Dual Match     | Lexical + Semantic Matchers   | Computes exact/alias/fuzzy token overlap (\$L\$) and dense cosine similarity (\$S\$) via all-MiniLM-L6-v2. |
| 4. Hybrid Score   | Requirement Scoring Engine    | Fuses lexical certainty with semantic context using closed-form dynamic weighting and noise cutoffs.       |
| 5. Ranking        | Deterministic Ranking Service | Monotonic sort with tie-breaks prioritizing required skill fulfillment, seniority, and score.              |
| 6. Explainability | Evidence-First Generator      | Stitches exact resume quote spans, lists matched concepts, and highlights missing mandatory skills.        |
| 7. Web Interface  | FastAPI Async + React 19 UI   | Recruiter dashboard with batch upload, live progress, shortlist cards, drawer, and pairwise comparison.    |

2. Mathematical Scoring Model & Recruiter Feature Suite

**Hybrid Scoring Formulation:**

- **Exact / Alias Lexical Match:**  $\text{Score} = 0.65 \cdot L + 0.35 \cdot S$  (proven tech usage + depth)
- **Fuzzy Lexical Match ( $L \geq 0.85$ ):**  $\text{Score} = 0.50 \cdot L + 0.50 \cdot S$  (spelling resilience)
- **Pure Conceptual Match:**  $\text{Score} = 0.75 \cdot S$  (e.g. *Express*  $\rightarrow$  *Node.js*; capped at 0.75)
- **Anti-Hallucination Cutoff:** If  $S < 0.35$ , semantic credit is set to 0.0 (noise rejection)
- **Priority Weighting:**  $\text{Final} = \Sigma(w_i \cdot \text{score}_i) / \Sigma w_i$  (Required: 3 $\times$ , Preferred: 1 $\times$ )

**Recruiter Tools & Rubric Innovations:**

- **Top-3 Auditable Briefs:** Natural-language summaries with verbatim provenance quotes.
- **Pairwise Comparison:** Answers "*Why is Candidate X above Y?*" via criteria delta matrix.
- **Ranking Robustness Audits:** Perturbation testing proves rank stability against synonym/formatting noise.
- **JD Bias Detection:** Flags exclusionary phrasing or narrow requirements in job postings.
- **Zero Cost & Offline:** Runs locally on CPU/GPU without cloud subscriptions or tokens.

3. Competition Evaluation Benchmarks & Technology Matrix

| Evaluation Metric    | Observed Benchmark Result                                   | System Layer  | Open-Source Technologies Applied                  |
|----------------------|-------------------------------------------------------------|---------------|---------------------------------------------------|
| Parse Success Rate   | 100% across 18 official + 54 external audit PDFs            | Backend API   | FastAPI, Uvicorn, Pydantic v2, Python 3.9+        |
| Score Distribution   | Strong: 85-94   Medium: 60-78   Weak: 25-45 (Clean spread)  | Parsing & NLP | PyMuPDF (fitz), Sentence-Transformers, PyTorch    |
| Robustness Stability | Top-3 candidate ranks invariant to formatting perturbations | Frontend UI   | React 19, Vite, Tailwind CSS, Lucide Icons        |
| Batch Throughput     | ~1.2s per 18-resume batch (Local CPU inference)             | Verification  | Pytest, Playwright UI tests, Perturbation Harness |

**Repository & Audit Verification:** All source code, scoring algorithms, test suites, and documentation are publicly available at <https://github.com/DreamX55/Nexora-Brocode.git>. Fully reproducible offline with pytest and HF\_HUB\_OFFLINE=1.